Active Control of a Laser-Actuated Photonic Metasurface

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Project Overview

Goal: Develop an active control scheme to stabilize and move a photonic metasurface which is levitated by a collimated laser beam.

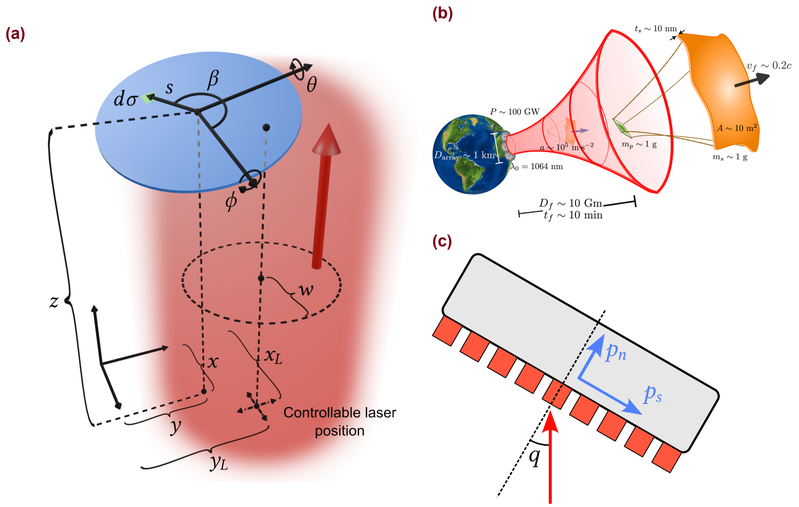


Figure 1. A collimated beam incident on the metasurface induces radiation pressure components $p_n(q, s)$ and $p_s(q, s)$ which govern surface dynamics. The laser position and intensity are used to control the surface position and orientation. (b) Concept art for implementing as interstellar lightsail (Lin 2025).

Metasurface Dynamics

Force and torque expressions are given by,

$$\mathbf{F}_{\text{body}} = I_0 \cos\theta \cos\phi \int_0^{2\pi} d\beta \int_0^{D/2} s\, ds \begin{bmatrix} p_s \cos\beta \\ p_s \sin\beta \\ p_n \end{bmatrix} I\left(\mathbf{r}_\perp\right) \tag{1}$$

$$\boldsymbol{\tau} = I_0 \cos\theta \cos\phi \int_0^{2\pi} d\beta \begin{bmatrix} \sin\beta \\ -\cos\beta \\ 0 \end{bmatrix} \int_0^{D/2} ds\, s^2 p_n\, I\left(\mathbf{r}_\perp\right) \tag{2}$$

with normalized beam profile $I(\mathbf{r}_\perp)$ given as a function of the distance from the beam axis $\mathbf{r}_\perp$, and scaled by max intensity I_0. The force vector is transformed from the surface body frame B to the fixed frame I via a rotation matrix $H_B^I(\phi, \theta, \psi)$.

Active Control Scheme

Dynamics cast as a first order ODE in the state and control vectors which may then be linearized about a fixed point,

$$\mathbf{x} = \begin{bmatrix} x\ y\ z\ \phi\ \theta\ \psi\ v_x\ v_y\ v_z\ \omega_1\ \omega_2\ \omega_3 \end{bmatrix}^T, \quad \mathbf{u} = \begin{bmatrix} x_L\ y_L\ I_0 \end{bmatrix}^T \tag{3}$$

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) \rightarrow \dot{\mathbf{x}} \approx A\Delta\mathbf{x} + B\Delta\mathbf{u} \tag{4}$$

Here $A = \left.\frac{df}{d\mathbf{x}}\right|_{\mathbf{x}^*,\mathbf{u}^*}$ and $B = \left.\frac{df}{d\mathbf{u}}\right|_{\mathbf{x}^*,\mathbf{u}^*}$ with $\mathbf{x}^*$ and $\mathbf{u}^*$ a fixed point wherein the metasurface is flat and unmoving, and the beam is centered on the metasurface. Typical observations of the system only include position,

$$\mathbf{y} = C\mathbf{x} = \begin{bmatrix} x\ y\ z \end{bmatrix}^T \tag{5}$$

A Kalman filter gives a dynamical equation to update the state estimate $\hat{\mathbf{x}}$, which we use to update our controller $\mathbf{u} = -K\hat{\mathbf{x}}$.

Simulation of Specular Reflector & Control Scheme

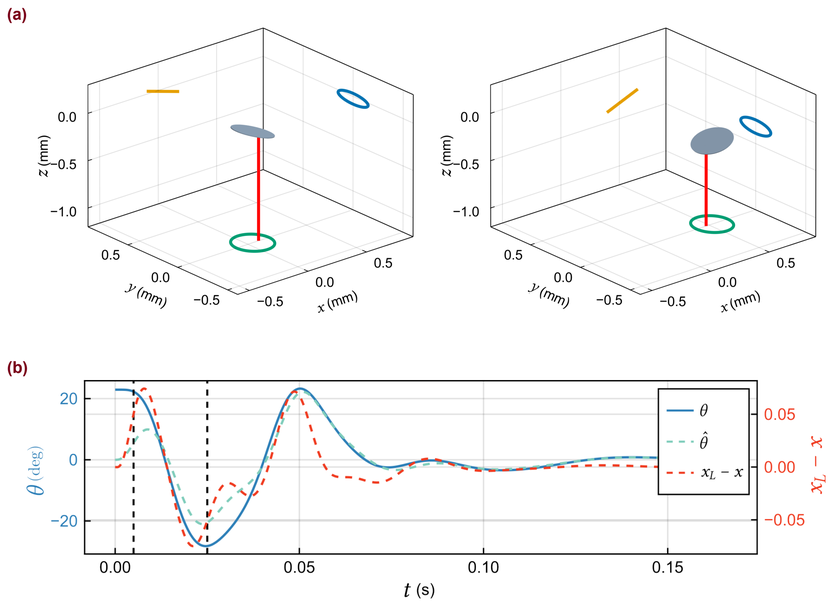


Figure 2. Trajectory of specular reflecting disc with $\theta_0 = 23°$. a) Snapshots of configuration, b) θ coordinates as a function of time. Vertical dashed lines are snapshot times.

Adapting Control Scheme to Experiment

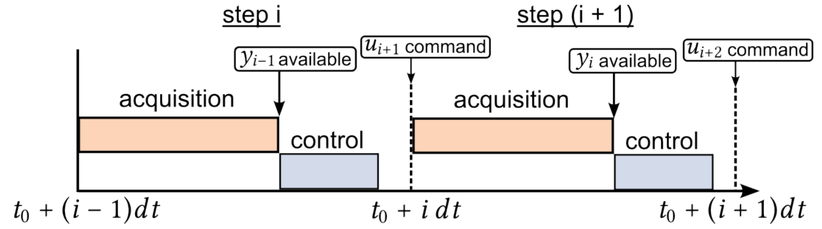


Figure 3. Experimental control loop with data acquisition and control steps.

Control loop for step i:

1. Use $\mathbf{y}_{i-1}$ and $\mathbf{u}_{i-1}$ to update $\hat{\mathbf{x}}_{i-2} \rightarrow \hat{\mathbf{x}}_{i-1}$.
2. Use the linear dynamics A, B and $\mathbf{u}_i$ to project forward $\hat{\mathbf{x}}_{i-1} \rightarrow \overline{\mathbf{x}}_i$.
3. Use to projected state vector to calculate control value $\mathbf{u}_{i+1} = -K\overline{\mathbf{x}}_i$.

All matrix operations, and may be done in real time.

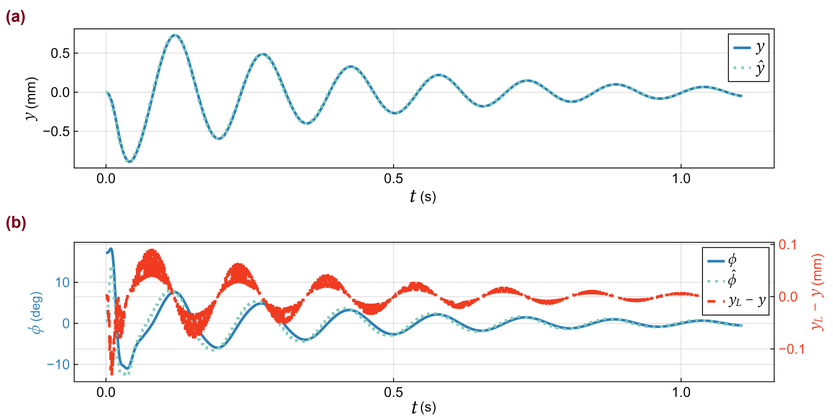


Figure 4. Trajectory with $\phi_0 = 17.2°$ and $dt = 2$ms.

References

"Optomechanical Self-Stability of Freestanding Photonic Metasurfaces" Kumar, Kindem, Ilic *Phys. Rev. Applied*, 16, 014053 (2021).

"Lightsails for Interstellar Travel: Photonics for Propulsion, Thermal Management and Stability" Lin, de Sterke, Ilic, Kuhlmey *ACS Photonics*, 12, 4818 - 4850 (2025).

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